



Afenich Amal

Engineering Student in Big Data & Artificial Intelligence

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PROFILE

Final-year Big Data & Artificial Intelligence engineering student at ENSA Tétouan, seeking a final-year project (PFE) internship starting January 2027 in this field. Rigorous and autonomous, I want to apply my skills in machine learning, data analysis and MLOps (Azure, Docker) to contribute to innovative projects.

EDUCATION

State Engineering Degree – Big Data & Artificial Intelligence

ENSA Tétouan

2024 – Present

Preparatory Cycle (2AP) – 2 preparatory years

ENSA Tétouan

2022 – 2024

Baccalaureate in Physical Sciences – Very Good honors

Lycée Med Ben El Hassan El Ouazzani – Khémisset

2022

PROFESSIONAL EXPERIENCE

PFA Internship: MLOps Pipeline on Azure Machine Learning

Smartovate

July – August 2026

- Automated the training pipeline of a regression model (Scikit-learn) with Azure Machine Learning
- Experiment tracking and model versioning with MLflow
- CI/CD with GitHub Actions and model deployment as a REST API via an Azure ML endpoint
- A/B testing between a champion model (80%) and a challenger (20%)

EXTRACURRICULAR ACTIVITIES

Active Member – AI Club, ENSA Tétouan

Organization of the Microsoft event in collaboration with the AI club of ENSA Tétouan.

Active Member – Mechatronics Club, ENSA Tétouan

Organization of the National Robotics Competition (CNR) within the mechatronics club of ENSA Tétouan.

Hackathons – Rabat and Martil

Participation in hackathons in Rabat and Martil, where I worked in a team to solve practical challenges within a limited time and present our solutions to a jury.

CERTIFICATIONS

AI & Machine Learning

- Introduction to Artificial Intelligence and its Applications
- Supervised Learning
- Unsupervised Learning

Data & BI

- SQL
- Oracle SQL
- Power BI
- Tableau

Cloud & DevOps

- Docker
- Azure

LANGUAGES

Arabic: Native • French: Intermediate • English: Intermediate

SKILLS

Machine Learning & AI

Machine Learning, Deep Learning, NLP.

Data & Visualization

Python, SQL & Oracle, Power BI, Tableau.

Containerization

Docker.

PROJECTS

Real-Time Weather Data Streaming Pipeline

- Real-time weather data collection via the OpenWeather API with Python
- Data streaming with Kafka and climate data processing with Spark
- Interactive visualization with Streamlit and climate prediction models

SignSense AI – American Sign Language (ASL) Recognition

- Preprocessing and annotation of gesture images (NumPy, Pandas)
- Training and evaluation of a CNN model with TensorFlow / Keras
- Real-time ASL gesture recognition via camera with OpenCV, converted to text and speech

BioSentry – Early Detection of Drug Side Effects

- Multi-source scrapers (DailyMed, MedlinePlus, Askapatient) with Python and Selenium, stored in MongoDB
- Web usage clustering with K-Means, Agglomerative Clustering and HDBSCAN
- Anomaly detection on aggregated data with MAD Z-score, ECOD and Isolation Forest